INTERVIEW PACK

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DURATION	TECHNICAL	BEHAVIORAL	RISKS FLAGGED
60 min	7 questions	5 questions	3

01 EXECUTIVE SUMMARY

Shrijal Goswami, ML Engineer Intern candidate with ~1.5 years of hands-on ML and full-stack AI product experience

WHY SHORTLISTED Demonstrated end-to-end ML pipeline development, hybrid retrieval/RAG systems, production deployment via FastAPI/Docker, strong quantitative impact, and top ATS ranking for the NovaScale AI internship

DIFFERENTIATORS

- Built a hybrid BM25 + Sentence-Transformer retrieval pipeline achieving MRR 0.84 and ~45 ms latency
- Reduced LLM token cost per booklet by ~10× through prompt versioning and staged generation
- Delivered multiple production-grade AI services (Resume Intelligence Platform, AKademyOS, HireLens) as sole engineer

02 INTERVIEW PLAN

INTERVIEWER Senior ML Engineer / ML Platform Engineer with experience in RAG, vector databases, and cloud-native model serving

#	STAGE	TIME	FOCUS
1	**Intro & Fit**	5 min	Candidate motivation, role expectations, cultural alignment
2	**Technical Deep-Dive**	30 min	ML fundamentals, project specifics, system design, deployment
3	**Behavioral Assessment**	15 min	Leadership, ownership, conflict resolution, communication
4	**Wrap-up & Questions**	5 min	Candidate queries, next steps

PRIORITY FOCUS

- Hybrid retrieval/RAG design
- Production deployment with FastAPI/Docker
- Prompt engineering & cost optimization
- Feature engineering & model evaluation
- System architecture & scalability

03 TECHNICAL QUESTIONS

T1 What is the difference between supervised and unsupervised learning, and can you give a concrete example of each you have worked on?

EASY · ML FUNDAMENTALS

WHY ASK Validate baseline understanding before probing deeper project work

STRONG ANSWER Supervised learning uses labeled data to predict targets (e.g., classification of heart disease using XGBoost). Unsupervised learning finds structure without labels (e.g., clustering resumes for similarity).

RED FLAGS

- Confuses definitions
- Cannot cite a real example

EVALUATE ON

- Clarity of definitions
- Relevance of examples
- Correct metric awareness

FOLLOW-UPS

- Which metric did you use for the supervised model?
- How did you evaluate the quality of the unsupervised clusters?

T2 Walk me through the hybrid retrieval pipeline (BM25 + Sentence-Transformer embeddings) you built for the Explainable RAG QA system. Include data ingestion, indexing, query processing, and how you combined sparse and dense scores.

MEDIUM | RETRIEVAL / RAG

WHY ASK
Directly ties to a high-impact project and tests depth of implementation knowledge

STRONG ANSWER
Ingestion parses documents, creates BM25 inverted index and dense embeddings via Sentence-Transformer, stores both in ChromaDB. At query time, BM25 returns top-k sparse scores, dense model returns similarity scores; final ranking is a weighted linear combination ($\alpha \approx 0.65$). Reranking and MRR evaluation follow.

RED FLAGS
- Omitting weighting scheme
- No mention of latency considerations

EVALUATE ON
- Understanding of both components
- Justification of weighting
- Awareness of performance trade-offs

FOLLOW-UPS
- How did you choose $\alpha=0.65$?
- What monitoring did you put in place for drift?

T3 Describe the feature engineering steps you performed for the Heart Disease Prediction stacking ensemble that achieved 0.94 ROC-AUC.

MEDIUM | FEATURE ENGINEERING & MODEL EVALUATION

WHY ASK
Assess practical data-preprocessing skills that are missing from the resume evidence

STRONG ANSWER
Handled missing values with median imputation, encoded categorical variables via one-hot, created interaction features (age × cholesterol), performed scaling for tree-based models, used cross-validation to select features, and stacked XGBoost, LightGBM (simulated) and logistic regression with meta-learner.

RED FLAGS
- Only generic statements
- No mention of specific features

EVALUATE ON
- Specificity of steps
- Awareness of leakage
- Metric-driven decisions

FOLLOW-UPS
- How did you prevent data leakage in stacking?
- Which metric guided your feature selection?

T4 Explain how you containerized the XGBoost model and exposed it via a FastAPI service, covering versioning, dependency management, and scaling considerations.

HARD | MODEL DEPLOYMENT (FASTAPI, DOCKER)

WHY ASK
Validate production-ready deployment experience claimed in multiple projects

STRONG ANSWER
Dockerfile based on python:3.10, installs requirements, copies model artifact, uses uvicorn to run FastAPI. Model versioning via path parameter or environment variable, stored in mounted volume. Scaling via Docker-Compose or Kubernetes replica set, health checks, and logging.

RED FLAGS
- No versioning strategy
- Missing health-check or scaling discussion

EVALUATE ON
- Correct Docker concepts
- Versioning approach
- Scalability awareness

FOLLOW-UPS
- How would you roll back to a previous model version?
- What monitoring would you add for latency?

T5 You reduced token cost per booklet by ~10× in the AKademyOS project. Detail the techniques you used (e.g., prompt versioning, staged generation, shared-context reuse) and how you measured the cost reduction.

HARD | PROMPT ENGINEERING & COST OPTIMIZATION

WHY ASK
Assess advanced LLM usage and quantitative impact evaluation

STRONG ANSWER
Implemented versioned prompt templates to reuse common context, staged generation separating outline from detailed content, cached embeddings for repeated sections, and measured token usage via OpenAI usage logs, confirming reduction from 40% to 4% of prior budget per booklet.

RED FLAGS
- Vague description of techniques
- No concrete measurement method

EVALUATE ON
- Depth of techniques
- Quantitative validation
- Awareness of quality-cost balance

FOLLOW-UPS
- Did you encounter any quality trade-offs?
- How would you further reduce cost without hurting accuracy?

T6 Your hybrid retrieval system currently yields MRR 0.84 with ~45 ms latency. Propose a systematic plan to push MRR to ≥0.90 while keeping latency ≤50 ms. Include model, indexing, and evaluation steps.

EXPERT | SYSTEM DESIGN & PERFORMANCE OPTIMIZATION

WHY ASK Test high-level problem-solving and ability to balance accuracy vs latency

STRONG ANSWER 1) Fine-tune dense encoder on domain data to improve relevance; 2) Adjust α weighting via grid search; 3) Introduce quantized embeddings (e.g., ONNX) to speed up distance calculations; 4) Use hierarchical IVF-PQ indexing for faster ANN; 5) Implement caching of frequent queries; 6) Re-evaluate with held-out set, monitor latency with profiling tools; 7) Iterate until target met.

RED FLAGS
- Ignoring latency constraint
- Suggesting heavyweight models without quantization

EVALUATE ON
- Feasibility of steps
- Balance of accuracy vs latency
- Awareness of bias & monitoring

FOLLOW-UPS
- How would you validate that the new model does not introduce bias?
- What fallback would you provide if latency spikes?

T7 You have not mentioned systematic experiment tracking in your resume. How would you integrate a tool like MLflow into your existing Resume Intelligence Platform pipeline to ensure reproducibility and auditability?

EXPERT | EXPERIMENT TRACKING & MLOPS

WHY ASK Address a clear gap in the candidate's skill set and gauge learning ability

STRONG ANSWER Add MLflow tracking server, wrap training scripts with mlflow.start_run(), log parameters (model hyper-params, data version), log metrics (accuracy, latency), log artifacts (model files, plots). Use conda environment for reproducibility, tag runs with Git commit SHA, and expose UI for audit. Integrate with CI/CD to auto-register new models.

RED FLAGS
- No mention of logging artifacts or version control linkage

EVALUATE ON
- Understanding of MLflow components
- Linkage to version control
- Operational considerations

FOLLOW-UPS
- How would you handle secret management for API keys in MLflow?
- What would you do if a run fails mid-training?

04 BEHAVIORAL QUESTIONS

B1 Describe a time when you led the design and delivery of an end-to-end AI product (e.g., AKademyOS). What was your role, how did you coordinate with others, and what was the outcome?

LEADERSHIP

WHY ASK Candidate claims sole engineer but also mentions team collaboration; probe real leadership experience

STRONG ANSWER Took ownership of architecture, defined milestones, held weekly syncs with front-end and dev-ops, delegated API contracts, delivered platform to 35+ users on schedule.

WARNING SIGNS
- No clear leadership actions
- Only individual contribution without coordination

B2 Tell me about a situation where you disagreed with a teammate on model selection for a project. How did you resolve the disagreement?

CONFLICT

WHY ASK Assess conflict resolution and data-driven decision making

STRONG ANSWER Presented validation results for both models, discussed trade-offs, reached consensus to use XGBoost for interpretability, documented decision.

WARNING SIGNS
- Avoided conflict, blamed others, no data-driven rationale

B3 Give an example where you identified a performance bottleneck in a production ML service and took initiative to fix it.

OWNERSHIP

WHY ASK Validate proactive ownership beyond stated responsibilities

STRONG ANSWER Observed high latency in the Resume Intelligence API, profiled request path, introduced caching layer with Redis, reduced latency by 60%, updated monitoring dashboards.

WARNING SIGNS
- Blames external factors, no concrete action

B4 When building the Resume Intelligence Platform, how did you decide between using BM25 versus dense embeddings for retrieval?

DECISION-MAKING

WHY ASK Probe analytical reasoning behind key architectural choice

STRONG ANSWER Ran A/B experiments on relevance (MRR), measured latency, considered compute cost; chose hybrid approach with α=0.65 to balance precision and speed.

WARNING SIGNS
- Decision based on intuition only, no metrics

B5 Explain a complex ML concept you had to present to non-technical stakeholders (e.g., educators using AKademyOS). How did you ensure they understood? COMMUNICATION

WHY ASK	Assess ability to translate technical ideas for business users
STRONG ANSWER	Used analogies (e.g., recommendation engine as a librarian), visual dashboards, interactive demos, collected feedback, iterated explanations.
WARNING SIGNS	• Overly technical language, no stakeholder feedback loop

05 SKILL VERIFICATION

SKILL	HOW TO VERIFY	CONFIDENCE
Python	Live coding exercise EXERCISE Implement a simple scikit-learn pipeline that loads a CSV, performs one-hot encoding, trains a logistic regression, and outputs accuracy DISCUSS Pythonic coding style and error handling	High
Scikit-learn & XGBoost	Project deep-dive EXERCISE Explain hyperparameter tuning process you used for the heart disease model and walk through code snippets DISCUSS Model evaluation metrics and cross-validation	High
FastAPI & Docker	Design exercise EXERCISE Sketch a Dockerfile and FastAPI endpoint for serving a pretrained model, discuss versioning strategy DISCUSS Container orchestration basics	Medium
Hybrid Retrieval / RAG	System design whiteboard EXERCISE Design a retrieval pipeline that combines BM25 and dense vectors, specify data flow and latency considerations DISCUSS Vector DB choice and weighting	Medium
Git/GitHub	Version-control scenario EXERCISE Explain branching strategy for collaborative feature development and how you would handle a merge conflict DISCUSS CI/CD integration	High
Experiment Tracking	Conceptual discussion EXERCISE Outline steps to integrate MLflow into an existing training script DISCUSS Reproducibility and auditability	Low
Cloud Deployment (AWS/GCP/Azure)	Scenario question EXERCISE Describe how you would migrate a Dockerized FastAPI service to AWS ECS/EKS, including networking and secrets DISCUSS Cloud-native considerations	Low

06 RISK ASSESSMENT

Timeline inconsistency

SIGNAL	Education dates (Sep 2024-Oct 2028) overlap with full-time work experience, suggesting possible resume inflation or future-dated claims
INVESTIGATE	Ask for current academic status, transcripts, and clarify how work was balanced with studies

Tooling gaps

SIGNAL	No explicit evidence of NumPy/Pandas usage or systematic experiment-tracking tools despite ML focus
INVESTIGATE	Probe concrete data-wrangling examples and ask about experiment tracking workflow in past projects

Limited cloud platform exposure

SIGNAL	Deployment experience limited to containers; no proven experience with major cloud services (AWS/GCP/Azure) required for scaling
INVESTIGATE	Discuss past cloud deployment scenarios, willingness to learn, and any informal exposure